Day 1 – Multi-Phase Pipeline Challenge (Debug → Validate → Extend → Optimize → Explain)

Objective

Move from writing queries to thinking like a data engineer. You will diagnose a broken pipeline, fix it, prove correctness, extend it with business logic, optimize it, and explain your decisions.

Dataset

Use the provided starter file (dirty data). Tables include customers and orders with issues such as nulls, invalid foreign keys, negative values, and inconsistent strings.

Expected Effort

7–8 hours (split into phases below). Work in steps. Do not jump ahead.

Phase 1 – Debug & Fix (1.5–2 hrs)

Goal: Identify why the current pipeline is wrong and fix it.

Tasks:

- Run the given pipeline and inspect outputs.
- Print schema and basic counts.
- Identify issues (nulls, negative values, invalid customer_id, duplicates, trimming).
- Fix data: handle nulls, filter invalid values, standardize types and strings.
- Re-run pipeline with corrected logic.

Deliverables: cleaned DataFrame(s) + corrected pipeline output.

Phase 2 – Validation & Trust (1.5–2 hrs)

Goal: Prove that your output is correct.

Tasks:

- Compute total_records, invalid_records, clean_records.
- Use left_anti join to identify invalid customer_id records.
- Compare counts before and after cleaning.
- Create a small validation report table.

Deliverables: validation_report_df with counts and checks.

Phase 3 – Extend the Pipeline (1.5–2 hrs)

Goal: Add business insights beyond the original pipeline.

Tasks (choose at least 3):

- Customer lifetime value (total spend per customer).
- City-wise contribution percentage to total revenue.
- Identify repeat customers (>2 orders).
- Detect anomalies (unusually high spend).
- Monthly trends using date functions.

Deliverables: aggregated_df / insights_df.

Phase 4 – Constraints & Optimization (1–1.5 hrs)

Goal: Improve efficiency and structure.

Constraints (pick one):

- Minimize number of groupBy operations.
- Reuse intermediate DataFrames (avoid recomputation).
- Use caching where beneficial.
- Avoid unnecessary wide transformations.

Deliverables: optimized pipeline with brief notes.

Phase 5 – Explain & Present (1 hr)

Goal: Communicate your thinking clearly.

README must include:

- What was wrong in the original pipeline?
- What steps did you take to fix it?
- How did you validate correctness?
- What assumptions did you make?
- Key learnings.

Output

Write final output to /tmp/day1_output and include screenshots in outputs/ folder with explanations.

Git Structure

Day1_Pipeline/ → README.md, pipeline.py, outputs/